

Degree program SYND	Academic year 2025-26	Nbr Thesis IT/2026/10
Mandant ☒ HES—SO Valais-Wallis ☐ Industry ☐ Partner institution	Student Gomes Costa Paulo Junior Professor Steiner Rico	Place of execution ☒ HES—SO Valais-Wallis ☐ Industry ☐ Partner institution
Confidential work ☐ Yes ☒ No	Expert (full name, E-Mail, Address) Beauge Gil	

Title

Development of a test probe as part of an generic test suite for performing distributed functional and security-related tests in OT environments.

Description

Current standards for OT environments require numerous functional and security-related tests, which must be defined as well as repeated and logged each time the system is modified. Preparing the test environment and conducting these tests is a complex procedure, as manipulations must be performed at various points of the system and the effects need to be observed.

To solve this problem, a system consisting of two parts comes into play:

- Probes in the form of an embedded system are to be developed. Placing these probes in the test environment allows to perform test operations such as but not limited to ping, HTTP requests, capturing Ethernet traces, reading and writing I/Os, etc.
- A central test management and coordination system (coordinator), which is NOT part of this project, performs predefined tests by interacting with the probes.

Having such a system allows automation of complex test procedures, with reproducible and comparable results.

The probe shall provide the following functions:

- Getting registered to a coordinator
- Sending the list of capabilities to the coordinator
- Receiving commands from the coordinator, executing them and sending the results back to the coordinator
- Local logging of the executed commands

Individual project (Pr4) objectives

- Study and comparison of existing, similar projects or products (commercial and/or open source).
- Evaluation of possible system architectures
- Defining the technologies, tools, libraries, and hardware platform for realizing the project
- Purchasing of the hardware as well as conducting a fundamental proof of concept
- Defining the interface (capabilities, technologies & protocol) between the probe to be developed and the management system
- Defining the scope of the bachelor thesis and elaboration of the system specification
- Establish a detailed Gantt diagram and work plan for the Bachelor thesis, including milestones and deliverables.

Bachelor's thesis (TB) objectives

- Documentation: produce the complete bachelor thesis report, including system architecture, experiments, evaluation results, and user/technical documentation.
- Design and implementation of the probe according to the specifications and results derived from PR4.
- Performing functional, performance and stability tests
- Building of a sample test environment for demonstrating the system's capabilities
- Optional: Integrate the work with a coordinator system and demonstrate interoperability of probes and coordinator.

Deadlines

Start of individual project (Pr4)

16.02.2026

Submission of the Pr4 report

01.05.2026, 12:00

Presentation of the individual project (Pr4)

Week 19 (04-08.05.2026)

Start of bachelor's thesis

18.05.2026

Submission of the final report

14.08.2026, 12:00

Exhibitions and pitch of bachelor's theses

28.08.2026 – HEI

31.08.2026 – Monthey

03.09.2026 – Visp

Oral defense

Week 35-37 (24.08-11.09.2026)

Signature or visa

Head of the major



¹ Student



¹ By signing, the student agrees to comply with guideline DI.1.2.02.07 "Bachelor's Thesis".